

Computer Science Ph.D. Candidate building reliable machine-learning systems for network security, computer vision, and multimodal AI. Thirteen first-author works span lightweight DDoS detection, noise-tolerant and data-efficient learning, LLM-enhanced network analysis, foreground-centric recognition, and vision–language generation. Open to Research Scientist, Applied Scientist, and Machine Learning Engineer roles beginning Fall 2027.

TECHNICAL STRENGTHS

Languages	Python · C++
ML frameworks	PyTorch · TensorFlow · Ultralytics YOLO11 · scikit-learn
Engineering	Docker · OpenCV · reproducible training pipelines · model evaluation
Research	Network security · deep learning · LLM-enhanced analysis · computer vision · multimodal learning

FEATURED SYSTEM

warbler_yolo · End-to-end fine-grained vision pipeline

- Built a reusable command-line workflow that prepares unannotated field images, segments subjects with YOLO11-Seg, creates standardized foreground crops and data splits, and trains and evaluates a YOLO11-CIs model.
- Designed the pipeline to reduce background shortcuts in fine-grained recognition and to run across macOS, Linux, and Windows.
- Released the implementation supporting a peer-reviewed foreground-centric recognition study.

Python · PyTorch · Ultralytics YOLO11 · OpenCV · scikit-learn · Paper

DOCTORAL RESEARCH · FORDHAM UNIVERSITY

Ph.D. Expected 2027	Robust AI for network security and intelligent systems - Developed compact flow-level and noise-tolerant neural models for DDoS detection in edge, IoT, and resource-constrained network settings. - Investigated few-shot adaptation, synthetic augmentation, multi-model decision fusion, and LLM-enhanced traffic analysis for changing attacks and limited data. - Extended the same reliability focus to fine-grained vision and vision–language generation, connecting rigorous experiments with reproducible software.
------------------------	--

SELECTED PEER-REVIEWED WORK

- ShallowNet: A Lightweight Neural Network Approach for Efficient Flow-Level DDoS Detection.** IEEE Transactions on Network and Service Management, 2026. DOI
- A Two-Stage LLM-Enhanced DDoS Detection Framework for Next-Generation IoT and Edge Networks.** IEEE GLOBECOM, 2025. DOI
- Foreground-Centric learning improves robustness in fine-grained visual recognition.** Signal, Image and Video Processing, 2025. DOI
- Vision-Language Integration for Image Captioning Using Vision Transformers and GPT-J.** AIPR/LNCS, 2026. DOI

ADDITIONAL EXPERIENCE

Instructor · Fordham University Structures of Computer Science · Fall 2025 and Spring 2026.	Inventor · CN113537358B Cancer subtype identification via multi-omics data integration. Patent.
---	---